Vertical shifts and dilations of sine and cosine functions (Extended)

ANSWER KEY

Core Concepts: Vertical shifts and dilations of sine and cosine functions (Extended)

1 For the model $f(x) = 6x + 7$ in vertical shifts and dilations of sine and cosine functions (extended), evaluate $f(2)$.

a. 19 ✓

b. 25

c. 12

d. 23

Substitute $x = 2$: $f(2) = 6(2) + 7 = 19$.

2 Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

3 A data set recorded for vertical shifts and dilations of sine and cosine functions (extended) consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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Procedural Fluency & Algebraic Manipulation

4 For the model $f(x) = 6x + 7$ in vertical shifts and dilations of sine and cosine functions (extended), evaluate $f(3)$.

a. 25 ✓

b. 31

c. 18

d. 29

Substitute $x = 3$: $f(3) = 6(3) + 7 = 25$.

5 Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

6 A data set recorded for vertical shifts and dilations of sine and cosine functions (extended) consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

Graphical & Geometric Interpretation

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7 For the model $f(x) = 6x + 7$ in vertical shifts and dilations of sine and cosine functions (extended), evaluate $f(4)$.

a. 31 ✓

b. 37

c. 24

d. 35

Substitute $x = 4$: $f(4) = 6(4) + 7 = 31$.

8 Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

9 A data set recorded for vertical shifts and dilations of sine and cosine functions (extended) consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 6x + 7$ in vertical shifts and dilations of sine and cosine functions (extended), evaluate $f(5)$.

a. 37 ✓

b. 43

c. 30

d. 41

Substitute $x = 5$: $f(5) = 6(5) + 7 = 37$.

11 Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

12 A data set recorded for vertical shifts and dilations of sine and cosine functions (extended) consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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